

ESAT Physics

Comprehensive Revision Guide

With Tom R. — lesson by Alex Kealy (PMT Education)

Subject: Edexcel International A-Level (IAL) Physics
Focus: ESAT (Engineering & Science Admissions Test) preparation
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Scope of this guide. This guide consolidates a single ESAT physics tutorial covering a broad spread of admissions-style multiple-choice topics: the **Cosmic Microwave Background** and cosmological redshift, **radioactive decay chains**, **thermal conduction**, **graph interpretation** of mechanics relationships, **DC circuits** (maximum power, potential dividers, parallel branches, Kirchhoff's laws), **stellar luminosity scaling**, **wave/particle motion**, and **pulley dynamics**. Every worked example is drawn directly from the lesson. The guide is designed for fast, exam-focused revision under time pressure.

The ESAT rewards fast, accurate application of GCSE/AS-level physics. Speed matters as much as method — practise until each technique is automatic.

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1 Specification & Source Register

This guide was compiled from the sources listed below. Where a source was unavailable at compile time, that is recorded honestly so nothing is misrepresented as covered when it was not.

Source	Status	Contribution / Notes
Loom transcript	Used	Primary source. Full tutorial (tutor Alex Kealy) covering all topics in this guide. Lesson date 2026-07-11.
OneNote export	Missing	Not yet exported — flagged for a manual/cron export in a later pass. No content included from it.
SaveMyExams	Missing	No cached match for these topics. Not used.

Topic map (Edexcel IAL / ESAT overlap). The lesson deliberately ranged across units to mirror the ESAT's mixed multiple-choice style:

- **Astrophysics / cosmology** — CMBR, cosmological redshift, stellar luminosity.
- **Nuclear physics** — $\alpha/\beta/\gamma$ decay, balancing nuclear equations.
- **Thermal physics** — rate of thermal conduction.
- **Mechanics** — graph relationships, Newton's second law, pulleys.
- **Electricity** — power, potential dividers, parallel circuits, Kirchhoff's laws.
- **Waves** — particle motion, frequency, distance travelled.

2 Executive Summary — If You Remember Nothing Else

Core formulas

Wave speed: $c = f\lambda$ (c constant for light)

Ohm's law: $V = IR$

Electrical power: $P = IV = I^2R = \frac{V^2}{R}$

Kirchhoff 1 (current): $\sum I_{\text{in}} = \sum I_{\text{out}}$

Kirchhoff 2 (voltage): $\sum \varepsilon = \sum IR$ (closed loop)

Newton's 2nd law: $F_{\text{res}} = ma$

Kinematics: $a = \frac{\Delta v}{\Delta t}$, $v = u + at$

Work / energy: $W = Fd$, $E_p = mgh$, $E_k = \frac{1}{2}mv^2$

Thermal conduction rate: $C \propto \frac{A \Delta T}{L}$

Stellar luminosity: $L \propto r^2 T^4$ (Stefan-Boltzmann)

Key definitions & facts

- **Alpha particle** = ${}^4_2\text{He}$ (helium-4 nucleus).
- **Beta particle** = ${}^0_{-1}e$ (an electron).
- **Gamma** = ${}^0_0\gamma$ (high-energy EM radiation, no charge/mass change).
- **CMBR**: gamma radiation from the early universe, *stretched* (redshifted) into the microwave region by cosmic expansion — direct evidence of the Big Bang.
- **Potential divider**: series resistors split the supply voltage *in proportion to their resistances*.
- **Same wire** $\Rightarrow$ **same potential**: any points joined only by (ideal) wire sit at the same voltage.
- In one full oscillation a wave particle travels $4\times$ **the amplitude**.

Golden rules for the ESAT

1. **Resultant force** = ma , not just the applied force. Balance weight *plus* the ma term.
2. For **max power** in a *fixed* resistor, maximise current: make every *other* resistance minimum.
3. Choose the power form (I^2R vs V^2/R) that uses the quantity you already know is fixed.
4. “Minimum” / “maximum” in a question is chosen to make it *solvable* — use it as a hint.
5. A pulley redirecting a rope puts **twice** the tension on its coupling/axle.
6. Speed counts: get the method automatic so you can answer quickly under time pressure.

3 Core Theory from First Principles

3.1 Cosmological redshift and the CMBR

In the 1920s it was discovered that the universe is **expanding**. A direct consequence: the intense **gamma radiation** that filled the hot early universe has been *stretched* as space itself expanded, shifting it down through the spectrum into the **microwave** region. This relic radiation is the **Cosmic Microwave Background Radiation (CMBR)** — widely regarded as direct evidence of the Big Bang.

The physics is captured by the wave equation:

$$c = f\lambda.$$

Light always travels at c , so c is *constant*. If expansion **increases** the wavelength λ , then to keep $c = f\lambda$ constant the frequency f must **decrease**. Gamma (short λ , high f) $\rightarrow$ microwave (long λ , low f).

3.2 Radioactive decay and nuclear equations

Three decay modes, and what they do to the nucleus A_ZX (A = nucleon/mass number, Z = proton number):

Decay	Emitted particle	ΔA	ΔZ
Alpha (α)	${}^4_2\text{He}$	-4	-2
Beta-minus (β^-)	${}^0_{-1}e$	0	+1
Gamma (γ)	${}^0_0\gamma$	0	0

Conservation rule. In any nuclear equation the *total* nucleon number and the *total* proton number must balance on both sides. To test a candidate equation, add up A on each side, then add up Z on each side — if either fails to balance, the equation is wrong.

Building a chain. Suppose you must reduce the mass number by 4 while leaving the proton number unchanged. One α decay gives $A - 4$, $Z - 2$; two β^- decays give $Z + 2$ back. Net effect: $A - 4$, Z unchanged — exactly what was required. This “ α then 2β ” combination is a recurring ESAT pattern.

3.3 Rate of thermal conduction

The rate of thermal conduction C behaves like electrical conduction. It is:

- **inversely** proportional to length L (longer bar $\Rightarrow$ slower),
- **proportional** to cross-sectional area A (wider bar $\Rightarrow$ faster),
- **proportional** to the temperature difference $\Delta T = T_1 - T_2$.

$$C \propto \frac{A \Delta T}{L}$$

Why ΔT ? Energy flows because of the temperature *gradient*. If $\Delta T = 0$ there is no net flow. An **insulated** bar simply means no heat escapes through the sides — all the energy that enters one end leaves the other. To keep C constant while doubling A , you must also double L (since A/L is unchanged).

3.4 Interpreting mechanics graphs

Many ESAT questions ask which *physical situation* matches a given graph. The trick is to write the defining equation and read off shape / gradient:

Graph (y vs x)	Equation	Shape	Gradient means
GPE vs height	$E_p = mgh$	linear	mg
Work vs distance	$W = Fd$	linear	force F
Velocity vs time	$v = u + at$	linear	acceleration a
KE vs velocity	$E_k = \frac{1}{2}mv^2$	parabola	(not constant)

Key insight: **kinetic energy vs velocity is *not* linear** (it is quadratic), whereas GPE–height, work–distance and velocity–time are all straight lines through appropriate intercepts. The *gradient* of a work–distance graph is the force; of a velocity–time graph, the acceleration.

3.5 Electricity: power, dividers and parallel branches

Power — three equivalent forms. Starting from $P = IV$ and $V = IR$:

$$P = IV = I^2 R = \frac{V^2}{R}.$$

Pick the form that isolates the quantity that stays *fixed*. For maximum power in a *fixed* resistor R , since R cannot change you maximise the **current** through it — which means minimising every *other* resistance in series with it.

Potential divider. Series resistors share the supply voltage in the ratio of their resistances. Two resistors R_1, R_2 across supply V :

$$V_1 = \frac{R_1}{R_1 + R_2} V, \quad V_2 = \frac{R_2}{R_1 + R_2} V.$$

Everything joined by wire is at the same potential. The power supply fixes one side at (say) 12 V and the other at 0 V. Any points connected only by ideal wire *must* be at the same voltage: if there were a difference, free electrons would redistribute almost instantly to erase it. A potential *difference* can only appear across a component (a resistor), where charge does work. This single idea unlocks most parallel-circuit questions.

3.6 Kirchhoff's laws

The two laws

First law (current / charge conservation): the total current *into* any junction equals the total current *out*:

$$\sum I_{\text{in}} = \sum I_{\text{out}}.$$

Second law (voltage / energy conservation): around any closed loop the sum of EMFs equals the sum of potential drops:

$$\sum \varepsilon = \sum IR.$$

EMF is energy *given* to each unit charge; the charge spends all of it travelling once around the loop.

3.7 Stellar luminosity scaling (Stefan–Boltzmann)

A star's luminosity (power) obeys $L = \sigma AT^4$ with surface area $A \propto r^2$, so

$$L \propto r^2 T^4.$$

Scaling is then just a product of factors: multiply the radius by a factor k_r and the temperature by k_T and the luminosity scales by $k_r^2 k_T^4$.

3.8 Wave particle motion

A travelling wave has a sinusoidal shape, but each individual particle only oscillates **up and down** about its rest position — it does not move along with the wave. In **one full cycle** a particle goes down to the bottom, back up through the middle to the top, and back to rest: a total path length of **four amplitudes** ($4A$). Frequency f (in hertz) is the number of complete cycles per second, so the distance a particle travels per unit time is $4A \times f$.

3.9 Newton's second law with pulleys

The defining relation is $F_{\text{res}} = ma$: it is the *resultant* force that equals ma , not the applied force alone. For a mass lifted upward at acceleration a :

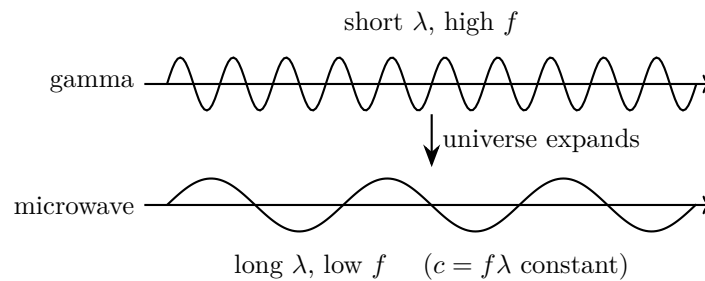
$$T - mg = ma \quad \implies \quad T = m(g + a).$$

A pulley or coupling that redirects the rope feels the tension on *both* sides pulling the same way, so the force on the coupling is $2T$.

4 Diagrams & Visual Layer

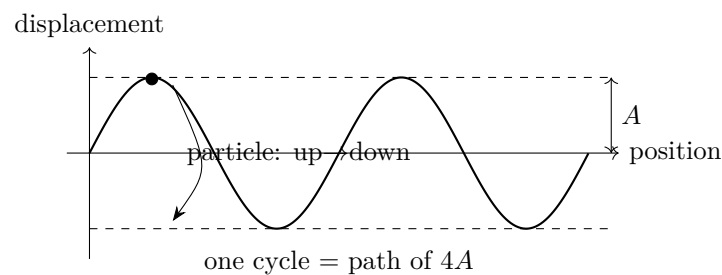
4.1 Cosmological redshift: stretching the wave

As space expands, a short-wavelength gamma wave is stretched into a long-wavelength microwave. The speed c is unchanged, so the frequency falls.



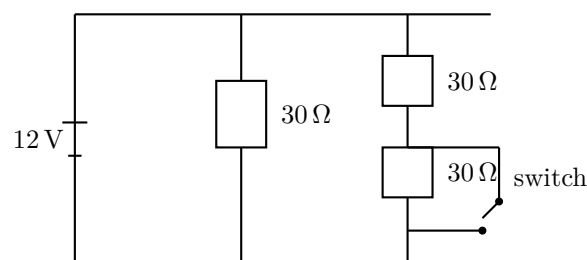
4.2 Wave particle motion — one full cycle

A single particle traces 4 amplitudes per cycle while the wave shape moves on.



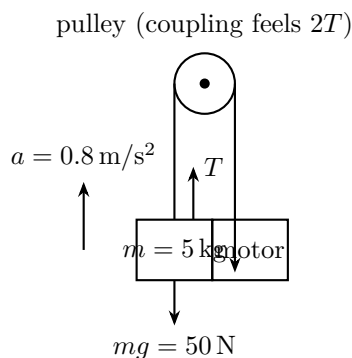
4.3 Parallel circuit with a switch (Worked Example 7)

Two parallel branches across a 12 V supply. Branch A: one $30\ \Omega$ resistor. Branch B: two $30\ \Omega$ resistors in series, with a switch that can short one of them out.



4.4 Motor–pulley system (Worked Example 10)

The motor pulls the rope down over a pulley; the tension lifts the mass. The pulley coupling supports the tension on *both* sides.



5 Worked Examples

All examples are taken directly from the lesson. Method, units and significant figures are shown in full.

5.1 WE1 — CMBR redshift (multiple choice)

The gamma radiation of the early universe is now observed as microwaves. What has happened to its frequency and wavelength?

Light speed is constant: $c = f\lambda$. Expansion **increases** λ . For c to stay constant, f must **decrease**.

$$\lambda \uparrow, \quad f \downarrow.$$

Answer: frequency decreased, wavelength increased (option B).

5.2 WE2 — Alpha/beta decay chain

A nucleus decays so that its mass number falls by 4 but its proton number is unchanged. Which sequence of decays achieves this?

Test the “ $1\alpha + 2\beta$ ” route:

$$\begin{aligned} \alpha : \quad A &\rightarrow A - 4, \quad Z \rightarrow Z - 2 \\ 2\beta^- : \quad Z - 2 &\rightarrow Z - 2 + 2 = Z \end{aligned}$$

Net: mass number $A - 4$, proton number *unchanged*. A route of three betas would raise Z by 3 (e.g. an incorrect $80 \rightarrow 83$), and a lone alpha lowers Z by 2 — neither matches. Only one alpha plus two betas works. **Answer: 1 alpha + 2 beta (option D).**

5.3 WE3 — Rate of thermal conduction

A copper bar and an insulator are arranged so the rate of conduction is unchanged. Which change to the temperatures keeps it constant?

Rate $C \propto A \Delta T / L$. The rate depends on the temperature *difference* ΔT , not the individual temperatures. If both ends rise by the same amount (e.g. both +10 K), then

$$\Delta T = T_1 - T_2 \text{ is unchanged } \Rightarrow C \text{ unchanged.}$$

Doubling A alone would *increase* C (rejected); raising only one temperature changes ΔT (rejected). **Answer: both temperatures raised equally (option H).**

5.4 WE4 — Matching graphs to mechanics

Which labelled straight-line graphs are physically consistent? Data: $g = 10 \text{ m/s}^2$.

(1) **GPE vs height**, point $(h, E_p) = (2 \text{ m}, 10 \text{ J})$, $m = 20 \text{ kg}$:

$$E_p = mgh = 20 \times 10 \times 2 = 400 \text{ J} \neq 10 \text{ J. (incorrect } \times)$$

(3) **Velocity vs time**, $m = 20 \text{ kg}$, gradient $= \frac{\Delta v}{\Delta t} = \frac{10}{2} = 5 \text{ m/s}^2$:

$$F = ma = 20 \times 5 = 100 \text{ N } \checkmark \text{ (matches label).}$$

(4) **Work vs distance**, gradient $= \frac{\Delta W}{\Delta d} = \frac{10}{2} = 5 \text{ N}$:

$$\text{gradient of } W-d \text{ graph} = F = 5 \text{ N } \checkmark \text{ (matches 5 N force).}$$

And KE vs velocity is a *parabola* ($E_k = \frac{1}{2}mv^2$), so any straight line for it is wrong. **Answer: graphs 3 and 4 are consistent.**

5.5 WE5 — Balancing a nuclear equation

Which candidate decay equation is balanced?

Rule: total nucleon number and total proton number must each balance. For the candidate with $A = 236$, $Z = 92$ (uranium region): add the nucleon numbers on each side, then the proton numbers. The valid equation has both sums equal on the two sides; the rejected one gives e.g. a bottom-row proton total that does not reconcile (92 vs an inconsistent value). **Answer: the equation whose A and Z both balance (the second option).**

5.6 WE6 — Maximum power in a fixed resistor

A 24 V supply drives a variable resistor (range 3 – higher) in series with a fixed 5Ω resistor. Find the maximum power dissipated in the 5Ω resistor.

$R = 5\ \Omega$ is fixed, so maximise the current: set the variable resistor to its **minimum**, $3\ \Omega$.
Total resistance:

$$R_{\text{tot}} = 3 + 5 = 8\ \Omega, \quad I = \frac{V}{R_{\text{tot}}} = \frac{24}{8} = 3\ \text{A}.$$

Power in the $5\ \Omega$ resistor (use $I^2 R$, since I is now known):

$$P = I^2 R = 3^2 \times 5 = 45\ \text{W}.$$

Check via potential divider: $V_5 = \frac{5}{8} \times 24 = 15\ \text{V}$, so $P = \frac{V^2}{R} = \frac{15^2}{5} = 45\ \text{W}$. ✓ **Answer:** 45 W.

Watch — the divider slip

[Phuc watch] In the lesson the current was first mis-computed as $24/5 = 4.8\ \text{A}$ by dropping the *whole* 24 V across the $5\ \Omega$ resistor. The supply is shared across *all* $8\ \Omega$: the correct current is $24/8 = 3\ \text{A}$. Always divide by the *total* series resistance.

5.7 WE7 — Parallel circuit, closing a switch

A 12 V battery feeds two parallel branches. Branch A: one $30\ \Omega$ resistor. Branch B: two $30\ \Omega$ resistors in series. A switch can short out the lower resistor in branch B. Find the current before and after closing the switch.

Branch A sees the full 12 V (nothing else in that branch):

$$I_A = \frac{V}{R} = \frac{12}{30} = 0.4\ \text{A}.$$

Branch B has two equal resistors, so the 12 V splits evenly: 6 V across each. Current:

$$I_B = \frac{6}{30} = 0.2\ \text{A} \quad \left(\text{equivalently } \frac{12}{60}\right).$$

Total (switch open):

$$I_0 = I_A + I_B = 0.4 + 0.2 = 0.6\ \text{A}.$$

Close the switch: the lower resistor is shorted, so branch B is now a single $30\ \Omega$ across the full 12 V:

$$I'_B = \frac{12}{30} = 0.4\ \text{A}, \quad I_{\text{new}} = 0.4 + 0.4 = 0.8\ \text{A}.$$

Answer: current rises from 0.6 A to 0.8 A (an increase of 0.2 A).

5.8 WE8 — Stellar luminosity scaling

A star's radius is multiplied by 100 and its temperature is halved. By what factor does its luminosity change?

Using $L \propto r^2 T^4$:

$$\text{radius factor: } (100)^2 = 10^4, \quad \text{temperature factor: } \left(\frac{1}{2}\right)^4 = \frac{1}{16}.$$

$$\text{Luminosity factor} = 10^4 \times \frac{1}{16} = \frac{10^4}{16} = 625.$$

Answer: luminosity increases by a factor of 625 (i.e. 6.25×10^2). No calculator needed — $10^4/16$ is just a quarter of 2500.

5.9 WE9 — Distance travelled by a wave particle

A wave has amplitude 3 cm and frequency 5 Hz. How far does a single particle travel in one minute?

Path length per cycle = $4A = 4 \times 3 = 12$ cm. Frequency 5 Hz = 5 cycles per second, so in one minute:

$$5 \times 60 = 300 \text{ cycles.}$$

Total distance:

$$300 \times 12 = 3600 \text{ cm} = 36 \text{ m.}$$

Answer: 3600 cm (36 m) per minute.

5.10 WE10 — Minimum tension in a pulley coupling

A motor lifts a 5 kg mass upward at 0.8 m/s^2 over a pulley. Find the minimum tension in the rope and the force on the pulley coupling. Take $g = 10 \text{ m/s}^2$.

Weight: $mg = 5 \times 10 = 50 \text{ N}$ (downward). Resultant force for the upward acceleration:

$$F_{\text{res}} = ma = 5 \times 0.8 = 4 \text{ N (upward).}$$

The tension must overcome the weight *and* supply the resultant:

$$T - mg = ma \Rightarrow T = m(g + a) = 5(10 + 0.8) = 54 \text{ N.}$$

The pulley coupling carries the tension on *both* sides of the rope, both pulling downward, so:

$$F_{\text{coupling}} = 2T = 2 \times 54 = 108 \text{ N.}$$

Answer: minimum tension 54 N; force on coupling 108 N.

Watch — why 54 not 50, and why 108 not 54

[Phuc watch] If the tension were only 50 N the mass would be in equilibrium and *not* accelerate. It must be 54 N to give the extra 4 N resultant. And because the pulley redirects the rope, the tension acts on the coupling from both sides $\Rightarrow$ the coupling feels $2T = 108 \text{ N}$. “Minimum” is used because an upward acceleration could in principle be any value; the minimum tension corresponds to the stated minimum acceleration.

6 Exam Technique

What the ESAT rewards

- **Speed under pressure.** You must get answers right *and* fast — there is heavy time pressure. Drill methods until they are automatic.
- **GCSE/AS foundations.** Many questions need little beyond solid GCSE physics (especially electricity equations) — but they are asked in a *hard* way. Master the fundamentals rather than chasing exotic content.
- **Mental arithmetic tricks.** Learn to simplify without a calculator: e.g. $24^2 = 2 \times 12^2$; $10^4/16 = 625$; recognise ratios like $5/8$.

Frequently-tested topics (tutor's observation)

Not officially published, but from many past papers the tutor notes:

- **Electricity** appears a lot — practise circuits heavily.
- **Efficiency** questions are a favourite.
- **Nuclear** comes up regularly, usually in a straightforward form.
- **Maximum-power** questions recur — learn the standard method.

Method discipline

1. **Read “min”/“max” as a hint.** Examiners pick the wording that makes the problem solvable — use it to decide your setup.
2. **Choose the right power form.** I^2R if current is fixed/known; V^2/R if voltage is fixed/known. Don't blindly start with $P = IV$.
3. **Always use the resultant force** in $F = ma$ — account for weight, friction and applied force before writing ma .
4. **Divide by total resistance**, not one resistor, when finding series current.
5. **Sanity-check magnitudes.** A current that looks “too high” (e.g. 4.8 A where you expected under 3 A) is a signal you mis-split a voltage.

Preparation resources. The ESAT is new (only a couple of years old), so the exam board recommends practising with the older **NSAA** papers, which are similar in style. A sensible plan: work through the physics sections of **NSAA 2017, 2018 and 2019**, timed, and review every mistake.

7 Common Traps & Phuc Watch-List

These are the specific slips that arose during the lesson, with corrections.

Trap 1 — Splitting voltage in a potential divider

[Phuc watch] Wrong: $I = 24/5 = 4.8$ A (dropping all 24 V across the $5\ \Omega$ resistor). **Right:** the supply spans the *total* $8\ \Omega$: $I = 24/8 = 3$ A; the $5\ \Omega$ resistor gets only $\frac{5}{8} \times 24 = 15$ V. *Fix:* always use total series resistance, then split voltages by resistance ratio.

Trap 2 — Forgetting the resultant force

[Phuc watch] **Wrong:** tension = weight = 50 N. **Right:** to accelerate upward the tension must exceed the weight by ma : $T = mg + ma = 54$ N. *Fix:* in $F = ma$ the F is the *resultant*, so add the ma term on top of balancing weight/friction.

Trap 3 — Pulley coupling force

[Phuc watch] **Wrong:** coupling force = $T = 54$ N. **Right:** a pulley carries the tension on *both* sides of the rope $\Rightarrow 2T = 108$ N. *Fix:* for a rope redirected over a pulley, double the tension for the axle/coupling.

Trap 4 — “KE vs velocity is linear”

[Phuc watch] **Wrong:** treating E_k - v as a straight line. **Right:** $E_k = \frac{1}{2}mv^2$ is a *parabola*. Only GPE–height, work–distance and velocity–time are linear here. *Fix:* write the equation first, then judge the shape.

Trap 5 — Confusing “same wire” with “same everything”

[Phuc watch] **Concept:** points joined by ideal wire share the *same potential*; potential *differences* appear only across components. Circuits feel hard until this clicks — the tutor stresses it is very commonly mistaught. *Fix:* label every wire-connected node with its voltage first, then work out drops across resistors.

Trap 6 — Decay-chain proton bookkeeping

[Phuc watch] **Wrong:** assuming three betas (would raise Z by 3, e.g. $80 \rightarrow 83$). **Right:** one alpha ($Z - 2$) plus two betas ($Z + 2$) leaves Z unchanged while A drops by 4. *Fix:* tally A and Z changes explicitly for each step.

8 Self-Test

Cover the answers. Mix of recall, calculation and explanation.

1. **(Recall)** What is the CMBR, and what does it provide evidence for?
2. **(Recall)** Give the symbol, mass number and proton number of an alpha and a beta particle.
3. **(Explain)** Using $c = f\lambda$, explain why cosmic expansion lowers the frequency of the relic radiation.
4. **(Calc)** A 6 V supply drives a $2\ \Omega$ variable resistor (minimum $2\ \Omega$) in series with a fixed $4\ \Omega$ resistor. Find the maximum power in the $4\ \Omega$ resistor.
5. **(Calc)** A star’s radius doubles and its temperature triples. By what factor does its luminosity change?
6. **(Calc)** A wave has amplitude 4 cm and frequency 2 Hz. How far does a particle travel in 10 s?
7. **(Calc)** A 2 kg mass is lifted at 1.5 m/s^2 over a pulley ($g = 10\text{ m/s}^2$). Find the rope tension and the force on the coupling.

8. **(Explain)** Why does closing a switch that shorts one resistor in a series branch *increase* the total current?
9. **(Recall)** State both of Kirchhoff's laws.
10. **(Explain)** Two ends of a conducting bar are each heated by 15 K. What happens to the rate of conduction, and why?

Answers

1. Relic gamma radiation of the early universe, stretched by expansion into the microwave region; direct evidence of the Big Bang.
2. Alpha = ${}^4_2\text{He}$ ($A = 4, Z = 2$); beta = ${}^0_{-1}e$ ($A = 0, Z = -1$).
3. c is constant; expansion raises λ , so $f = c/\lambda$ must fall.
4. $R_{\text{tot}} = 6\Omega$, $I = 6/6 = 1\text{ A}$, $P = I^2R = 1^2 \times 4 = 4\text{ W}$.
5. $L \propto r^2T^4$: factor = $2^2 \times 3^4 = 4 \times 81 = 324$.
6. Per cycle = $4A = 16\text{ cm}$; $2\text{ Hz} \times 10\text{ s} = 20\text{ cycles}$; $20 \times 16 = 320\text{ cm} = 3.2\text{ m}$.
7. $T = m(g + a) = 2(10 + 1.5) = 23\text{ N}$; coupling = $2T = 46\text{ N}$.
8. Shorting the resistor lowers that branch's resistance, so (at fixed supply voltage) its current rises, raising the total.
9. (1) $\sum I_{\text{in}} = \sum I_{\text{out}}$ at a junction; (2) $\sum \varepsilon = \sum IR$ around a closed loop.
10. Unchanged: $C \propto \Delta T$, and heating both ends equally leaves $\Delta T = T_1 - T_2$ the same.

9 Must-Memorise Appendix

9.1 Formulas not always given — know them cold

Quantity	Formula
Wave speed	$c = f\lambda$
Ohm's law	$V = IR$
Electrical power	$P = IV = I^2R = V^2/R$
Potential divider	$V_1 = \frac{R_1}{R_1 + R_2} V$
Kirchhoff 1	$\sum I_{\text{in}} = \sum I_{\text{out}}$
Kirchhoff 2	$\sum \varepsilon = \sum IR$
Newton's 2nd law	$F_{\text{res}} = ma$
Acceleration	$a = \Delta v / \Delta t$
Work done	$W = Fd$
Gravitational PE	$E_p = mgh$
Kinetic energy	$E_k = \frac{1}{2}mv^2$
Thermal conduction rate	$C \propto A\Delta T/L$
Stellar luminosity	$L \propto r^2T^4$
Wave particle path / cycle	$= 4A$
Pulley coupling force	$2T$

9.2 Definitions to memorise

- **CMBR:** redshifted relic radiation from the early universe, now in the microwave band; evidence for the Big Bang.
- **Alpha particle:** a helium-4 nucleus, ${}^4_2\text{He}$.
- **Beta-minus particle:** a fast electron, ${}^0_{-1}e$.

- **EMF:** energy supplied to each unit of charge by a source ($\text{J/C} = \text{V}$).
- **Potential difference:** energy transferred per unit charge across a component.
- **Frequency:** number of complete cycles per second ($1 \text{ Hz} = 1 \text{ cycle/s}$).
- **Amplitude:** maximum displacement of a particle from its rest position.

9.3 Units & constants

Quantity	Unit	Symbol
Current	ampere	A
Potential difference / EMF	volt	V
Resistance	ohm	Ω
Power	watt	W
Force	newton	N
Frequency	hertz	Hz
Energy / work	joule	J
Acceleration	metre per second ²	m/s^2
g (used in lesson)	10 m/s^2 (approx. Earth surface gravity)	

End of guide — prepared for Phuc Nguyen, ESAT physics preparation.
 Keep drilling electricity and mechanics; speed is as important as accuracy.